On the Use of a Convex Model for Bulk Storage in MIP-Based Power System Operation and Planning

José M. Arroyo, *Fellow, IEEE*, Luis Baringo, *Senior Member, IEEE*, Ana Baringo, Ricardo Bolaños, Natalia Alguacil, *Senior Member, IEEE*, and Noemi G. Cobos

***Abstract*—Recently, several authors have reported specific criteria to support the widely accepted use of a convex operational model for bulk energy storage in the formulation of power system operation and planning problems involving binary variables. Using two counterexamples, we show that this modeling simplification may give rise to impractical solutions featuring simultaneous charging and discharging, even though the recently published criteria are satisfied. Thus, both counterexamples invalidate those criteria and demonstrate that the customarily used convex model may be unsuitable for the precise incorporation of bulk storage in such instances of mixed-integer programming.**

***Index Terms*—Convex operational model, electrical energy storage, power system operation and planning with binary variables, simultaneous charging and discharging.**

I. Introduction

ELECTRICAL energy storage devices are increasingly being incorporated into the optimization models used for power system operation [1]–[5] and planning [6]–[8]. The effective characterization of storage units requires accounting for the mutually exclusive operational states featured by these devices, namely charging, discharging, and idle. This modeling aspect can be implemented by nonconvex formulations, e.g., through either additional binary variables [5] or bilinear terms involving the product of two continuous variables [3], thereby increasing the mathematical complexity of the resulting optimization problems.

In [3], [6], the authors showed that a relaxed storage operational model can be equivalently adopted for two particular instances of power system operation and planning, respectively, wherein all decision variables are continuous and the terms in the objective function meet specific requirements.

Manuscript received April 30, 2020; revised July 13, 2020; accepted August 22, 2020. Date of publication September 1, 2020; date of current version November 4, 2020. This work was supported by the Ministry of Science, Innovation and Universities of Spain, under Projects RTI2018-098703-B-I00 and RTI2018-096108-A-I00 (MCIU/AEI/FEDER, UE). Paper no. PESL-00132-2020. *(Corresponding author: José M. Arroyo.)*

José M. Arroyo, Luis Baringo, Ana Baringo, and Natalia Alguacil are with the Departamento de Ingeniería Eléctrica, Electrónica, Automática y Comunicaciones, E.T.S.I. Industrial, Universidad de Castilla-La Mancha, Ciudad Real 13071, Spain (e-mail: josemanuel.arroyo@uclm.es; Luis.Baringo@uclm.es; Ana.Baringo@uclm.es; natalia.alguacil@uclm.es).

Ricardo Bolaños is with XM, Medellín 50022, Colombia (e-mail: rabolanos@gmail.com).

Noemi G. Cobos is with the Instituto Tecnológico de la Energía, Valencia 46980, Spain (e-mail: noemigonzc@gmail.com).

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Digital Object Identifier 10.1109/TPWRS.2020.3020730

Unfortunately, there is wide acceptance in the power system community that a convex formulation for storage operation can also be adopted in more complex operational and planning models involving binary variables such as those related to generation scheduling and investment decisions. References [2] and [7] are relevant examples wherein even specific criteria were provided to support the adoption of such a simplified model under a mixed-integer programming (MIP) framework. Within this MIP-based context, a convex storage operational model was also adopted, albeit without any sound justification, in [1], [4], [8], thereby backing its customary use in the power system literature.

In this letter, we give two counterexamples to convince the reader that the widely accepted convex model may fail to precisely represent storage operation, even though the conditions mentioned in [2] and [7] are satisfied. To that end, we provide the formulation and test results for two simple instances of practical MIP-based problems. The first model is a unit commitment problem that is useful to disprove the criterion argued in [2]. The second model is an expansion planning problem that invalidates the claim stated in [7]. Both models are simpler to describe and analyze, yet bringing out the issue of adopting a convex formulation for storage operation. Note that this issue also arises in more complex instances such as those addressed in [1], [2], [4], [5], [7], [8].

The main contributions of this letter are twofold:

1) To demonstrate via counterexamples the insufficiency of the conditions set forth in [2] and [7].
2) To raise awareness of the potential inadequacy of a convex model to precisely represent the operation of storage devices in MIP-based problems.

The significance of both contributions is backed by 1) the practical interest of operational and planning tools adequately representing system components, which may prevent costly corrective actions, thereby having important economic implications, and 2) the mathematical challenges posed by the resulting optimization problems, which render the methodologies reported in the literature for simpler models unsuitable for such more accurate formulations.

II. Convex Model and Supporting Conditions

The widely used convex model for storage operation comprises the following constraints:

$$e^{\mathrm{S}}_{st} = e^{\mathrm{S}}_{st-1} + \left(\eta^{\mathrm{S,C}}_{s} p^{\mathrm{S,C}}_{st} - \frac{1}{\eta^{\mathrm{S,D}}_{s}} p^{\mathrm{S,D}}_{st}\right) \Delta t; \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \quad (1)$$

$$\underline{E}^{\mathrm{S}}_{s} \leq e^{\mathrm{S}}_{st} \leq \overline{E}^{\mathrm{S}}_{s}; \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \quad (2)$$

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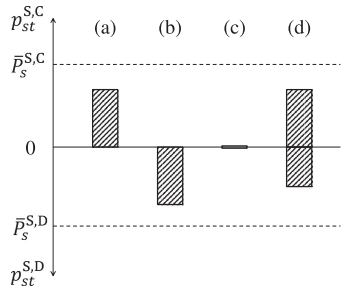


Fig. 1. Possible values of $p_{st}^{\mathrm{S,C}}$ and $p_{st}^{\mathrm{S,D}}$ in the convex model.

$$0 \leq p_{st}^{\mathrm{S,D}} \leq \overline{P}_s^{\mathrm{S,D}}; \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \tag{3}$$

$$0 \leq p_{st}^{\mathrm{S,C}} \leq \overline{P}_s^{\mathrm{S,C}}; \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \tag{4}$$

where storage units and time periods are respectively indexed by $s \in \mathcal{S}$ and $t \in \mathcal{T}$, whereas Δt denotes the time period duration. Decision variables include e_{st}^{S}, $p_{st}^{\mathrm{S,C}}$, and $p_{st}^{\mathrm{S,D}}$, respectively denoting the stored energy level, the charging power level, and the discharging power level of storage unit s in period t. In addition, $\eta_s^{\mathrm{S,C}}$ and $\eta_s^{\mathrm{S,D}}$ represent the charging and discharging efficiency rates of storage unit s, respectively, $\underline{E}_s^{\mathrm{S}}$ and $\overline{E}_s^{\mathrm{S}}$ are the minimum and maximum energy storage limits of storage unit s, respectively, whereas $\overline{P}_s^{\mathrm{S,C}}$ and $\overline{P}_s^{\mathrm{S,D}}$ respectively denote the rated charging and discharging power capacities of storage unit s.

The storage dynamics are cast in (1). Expressions (2) model the energy storage limits. Finally, the bounds on discharging and charging are respectively set in (3) and (4). Moreover, storage operation is typically characterized by linear charging and discharging costs, namely $C_s^{\mathrm{S,C}} p_{st}^{\mathrm{S,C}} \Delta t$ and $C_s^{\mathrm{S,D}} p_{st}^{\mathrm{S,D}} \Delta t$, where $C_s^{\mathrm{S,C}}$ and $C_s^{\mathrm{S,D}}$ respectively represent the charging and discharging cost coefficients of storage unit s.

As shown in Fig. 1, for storage unit s and period t, the above convex model is characterized by a feasibility space comprising the following four possible combinations of values for variables $p_{st}^{\mathrm{S,C}}$ and $p_{st}^{\mathrm{S,D}}$: 1) $p_{st}^{\mathrm{S,C}} > 0$ and $p_{st}^{\mathrm{S,D}} = 0$ (Fig. 1(a)), which corresponds to the charging state, 2) $p_{st}^{\mathrm{S,C}} = 0$ and $p_{st}^{\mathrm{S,D}} > 0$ (Fig. 1(b)), which corresponds to the discharging state, 3) $p_{st}^{\mathrm{S,C}} = 0$ and $p_{st}^{\mathrm{S,D}} = 0$ (Fig. 1(c)), which corresponds to the idle state, and 4) $p_{st}^{\mathrm{S,C}} > 0$ and $p_{st}^{\mathrm{S,D}} > 0$ (Fig. 1(d)), which does not correspond to any practical operational state, as charging and discharging in the same time period cannot be physically implemented.

In [2], the authors presented a condition to prevent the fourth combination from being featured at the optimum of the contingency-constrained unit commitment model addressed therein. A different condition was claimed in [7] to support the suitability of the convex model for planning with discrete line investment decisions. Both conditions are reproduced below:

- Wen *et al.*'s condition for the contingency-constrained unit commitment problem [2]: The presence of pre-contingency operating costs of storage units in the objective function.
- Zhao *et al.*'s condition for the transmission expansion planning problem [7]: A value for the roundtrip efficiency smaller than unity, i.e., $\eta_s^{\mathrm{S,D}} \eta_s^{\mathrm{S,C}} < 1$.

Unfortunately, the following counterexamples show that such conditions are not sufficient within an MIP-based framework. As numerically evidenced, if the above convex model is used, the cost increase associated with charging and discharging in the same time period may be offset by the reduction in other costs. Moreover, the convex model may also lead to simultaneous charging and discharging if such an impractical operational state allows artificially and economically complying with operational limits that would otherwise be violated.

III. First Counterexample: Unit Commitment

The first counterexample relies on a unit commitment model considering a single operating condition or base case. Thus, the optimization is solely driven by base-case cost terms, which meets the criterion set forth in [2] to disregard binary variables preventing the simultaneous charging and discharging of storage devices. For expository purposes, start-up and shutdown costs, minimum up and down times, reserves, the effect of the transmission network, and uncertainty are neglected.

Based on [5], this unit commitment instance is cast as the following mixed-integer linear program, where generators, storage units, and time periods are respectively indexed by $g \in \mathcal{G}$, $s \in \mathcal{S}$, and $t \in \mathcal{T}$:

$$\min \sum_{t \in \mathcal{T}} \Bigg[\sum_{g \in \mathcal{G}} \left(C_g^{\mathrm{G},0} v_{gt}^{\mathrm{G}} + C_g^{\mathrm{G,P}} p_{gt}^{\mathrm{G}} \Delta t \right) + \sum_{s \in \mathcal{S}} \left(C_s^{\mathrm{S,D}} p_{st}^{\mathrm{S,D}} + C_s^{\mathrm{S,C}} p_{st}^{\mathrm{S,C}} \right) \Delta t \Bigg] \tag{5}$$

subject to:

$$\sum_{g \in \mathcal{G}} p_{gt}^{\mathrm{G}} + \sum_{s \in \mathcal{S}} p_{st}^{\mathrm{S,D}} = P_t^{\mathrm{D}} + \sum_{s \in \mathcal{S}} p_{st}^{\mathrm{S,C}}; \forall t \in \mathcal{T} \tag{6}$$

$$\underline{P}_g^{\mathrm{G}} v_{gt}^{\mathrm{G}} \leq p_{gt}^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G}} v_{gt}^{\mathrm{G}}; \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \tag{7}$$

$$v_{gt}^{\mathrm{G}} \in \{0,1\}; \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \tag{8}$$

$$p_{gt}^{\mathrm{G}} - p_{gt-1}^{\mathrm{G}} \leq \Delta t \left[R_g^{\mathrm{G,UP}} v_{gt-1}^{\mathrm{G}} + R_g^{\mathrm{G,SU}} (v_{gt}^{\mathrm{G}} - v_{gt-1}^{\mathrm{G}}) \right] + \overline{P}_g^{\mathrm{G}} (1 - v_{gt}^{\mathrm{G}}); \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \tag{9}$$

$$p_{gt-1}^{\mathrm{G}} - p_{gt}^{\mathrm{G}} \leq \Delta t \left[R_g^{\mathrm{G,DN}} v_{gt}^{\mathrm{G}} + R_g^{\mathrm{G,SD}} (v_{gt-1}^{\mathrm{G}} - v_{gt}^{\mathrm{G}}) \right] + \overline{P}_g^{\mathrm{G}} (1 - v_{gt-1}^{\mathrm{G}}); \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \tag{10}$$

$$e_{st}^{\mathrm{S}} = e_{st-1}^{\mathrm{S}} + \left(\eta_s^{\mathrm{S,C}} p_{st}^{\mathrm{S,C}} - \frac{1}{\eta_s^{\mathrm{S,D}}} p_{st}^{\mathrm{S,D}} \right) \Delta t; \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \tag{11}$$

$$\underline{E}_s^{\mathrm{S}} \leq e_{st}^{\mathrm{S}} \leq \overline{E}_s^{\mathrm{S}}; \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \tag{12}$$

$$0 \leq p_{st}^{\mathrm{S,D}} \leq \overline{P}_s^{\mathrm{S,D}} z_{st}^{\mathrm{S}}; \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \tag{13}$$

$$0 \leq p_{st}^{\mathrm{S,C}} \leq \overline{P}_s^{\mathrm{S,C}} (1 - z_{st}^{\mathrm{S}}); \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \tag{14}$$

$$z_{st}^{\mathrm{S}} \in \{0,1\}; \forall s \in \mathcal{S}, \forall t \in \mathcal{T}. \tag{15}$$

The optimization yields the optimal values of the binary variables associated with generation scheduling and storage operation, v_{gt}^{G} and z_{st}^{S}, as well as the optimal generation levels, p_{gt}^{G}, stored energy levels, e_{st}^{S}, and charging and discharging power levels, $p_{st}^{\mathrm{S,C}}$ and $p_{st}^{\mathrm{S,D}}$. The set of parameters comprises the time period duration, Δt; the consumption levels, P_t^{D}; the no-load and variable generation cost coefficients, $C_g^{\mathrm{G},0}$ and $C_g^{\mathrm{G,P}}$; the minimum and maximum generation limits, $\underline{P}_g^{\mathrm{G}}$ and $\overline{P}_g^{\mathrm{G}}$; the generation upward, downward, start-up, and shutdown ramp rates, $R_g^{\mathrm{G,UP}}$, $R_g^{\mathrm{G,DN}}$, $R_g^{\mathrm{G,SU}}$, and $R_g^{\mathrm{G,SD}}$; the charging and discharging cost coefficients, $C_s^{\mathrm{S,C}}$ and $C_s^{\mathrm{S,D}}$; the charging

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TABLE I
UNIT COMMITMENT – OPTIMAL RESULTS

	With z^{S}_{st}		Without z^{S}_{st}	
	Hour		Hour	
	1	2	1	2
p^{G}_{1t} (MW)	12.3	27.3	13.8	28.8
p^{G}_{2t} (MW)	0.0	2.4	0.0	0.0
$p^{\mathrm{S,C}}_{st}$ (MW)	2.3	0.0	5.8	0.0
$p^{\mathrm{S,D}}_{st}$ (MW)	0.0	6.3	2.0	7.2
e^{S}_{st} (MWh)	12.0	5.0	13.0	5.0
MP ($/MWh)	2.6	3.4	−1.0	7.0
Total cost ($)	173.2		130.3	

and discharging efficiency rates, $\eta^{\mathrm{S,C}}_s$ and $\eta^{\mathrm{S,D}}_s$; the minimum and maximum energy storage limits, $\underline{E}^{\mathrm{S}}_s$ and $\overline{E}^{\mathrm{S}}_s$; and the rated charging and discharging power capacities, $\overline{P}^{\mathrm{S,C}}_s$ and $\overline{P}^{\mathrm{S,D}}_s$.

The optimization goal is the minimization of the total cost (5), which comprises terms associated with generators, namely no-load and variable generation costs, and storage units, namely discharging and charging costs. Constraints (6) are the power balance equations. Generation limits are imposed in (7) while accounting for on/off statuses v^{G}_{gt}, which are binary as per (8). Generation upward and start-up ramp rates are imposed in (9) whereas downward and shutdown ramp rates are modeled in (10). Finally, storage-related expressions (11)–(14) correspond to (1)–(4), respectively. Note, however, that, in (13) and (14), charging and discharging simultaneously is explicitly prevented by variables z^{S}_{st}, which are binary as modeled in (15). Thus, the representation of storage operation is nonconvex.

According to [2], the optimal solution to problem (5)–(15) can be attained by replacing (11)–(15) with (1)–(4), i.e., by dropping binary variables z^{S}_{st}. Unfortunately, the resulting relaxed problem with a convex storage operational model is not equivalent, as illustrated with the following case study wherein two generators, labeled as 1 and 2, and a storage unit are scheduled over two hourly periods with demands respectively equal to 10 MW and 36 MW. Both generators are initially scheduled off and feature the same upper and lower generation limits, namely 50 MW and 2.4 MW, as well as identical ramp rates, which are all equal to 15 MW/h. Generation cost coefficients are $C^{\mathrm{G},0}_1 = \$0.5/\mathrm{h}$, $C^{\mathrm{G},0}_2 = \$5.0/\mathrm{h}$, $C^{\mathrm{G,P}}_1 = \$3.0/\mathrm{MWh}$, and $C^{\mathrm{G,P}}_2 = \$19.9/\mathrm{MWh}$. As for the storage unit, the initial level of stored energy amounts to 10 MWh whereas the lower and upper bounds for the levels of stored energy are 5 MWh and 13 MWh, respectively. Furthermore, the rated power capacities and the efficiency rates for charging and discharging are equal to 12 MW and 0.9, respectively. Finally, as done in [2], the charging and discharging cost coefficients are identical, both amounting to $0.1/MWh.

Using the branch-and-cut algorithm of CPLEX, problem (5)–(15) and its relaxed version have been both solved to optimality, i.e., the optimality gap of CPLEX was set at 0%. Table I summarizes the optimal results. As can be observed, the use of a convex storage operational model without binary variables z^{S}_{st} yields a cheaper solution because the more expensive generator 2 is not scheduled on. However, such a cost reduction is attained by simultaneously charging and discharging the storage unit in period 1, which is impractical, i.e., it cannot be physically implemented in practice. The negative marginal price (MP) in period 1 corroborates the findings of [3] regarding the lack of equivalence of a convex model for storage operation. Moreover, note that the generation schedule resulting from the relaxed model is an infeasible solution to problem (5)–(15). These results clearly invalidate the criterion presented in [2] and demonstrate that, in general, simultaneously charging and discharging needs to be explicitly prevented in the problem formulation.

IV. SECOND COUNTEREXAMPLE: TRANSMISSION EXPANSION PLANNING

The second counterexample is based on a transmission expansion planning (TEP) model. TEP aims to determine the transmission lines to be installed in a power system to minimize both investment and operational costs. For the sake of clarity, a simplified deterministic version of the two-stage problem addressed in [7] is analyzed in which lines are the only candidate assets and expansion decisions are made for a single future operating condition comprising several representative time periods.

Let us index representative periods, buses, generators, lines (existing and candidate), and storage units by $t \in \mathcal{T}$, $b \in \mathcal{B}$, $g \in \mathcal{G}$, $\ell \in \mathcal{L}$, and $s \in \mathcal{S}$, respectively. In addition, let $\tilde{\mathcal{L}}$ denote the set of indexes ℓ of candidate lines, and let $\mathcal{G}_b$ and $\mathcal{S}_b$ respectively denote the sets of indexes of generators and storage units located at bus b. Based on [7] and [8], the TEP problem is formulated as the following mixed-integer nonlinear program:

$$\min \sum_{\ell\in\tilde{\mathcal{L}}} C^{\mathrm{L}}_\ell y^{\mathrm{L}}_\ell + \sum_{t\in\mathcal{T}} \Delta t \Bigg[\sum_{g\in\mathcal{G}} C^{\mathrm{G,P}}_g p^{\mathrm{G}}_{gt} + \sum_{s\in\mathcal{S}} \left(C^{\mathrm{S,D}}_s p^{\mathrm{S,D}}_{st} + C^{\mathrm{S,C}}_s p^{\mathrm{S,C}}_{st} \right) \Bigg] \tag{16}$$

subject to:

$$y^{\mathrm{L}}_\ell \in \{0,1\}; \forall \ell \in \tilde{\mathcal{L}} \tag{17}$$

$$\sum_{g\in\mathcal{G}_b} p^{\mathrm{G}}_{gt} + \sum_{s\in\mathcal{S}_b} p^{\mathrm{S,D}}_{st} + \sum_{\ell|to(\ell)=b} p^{\mathrm{L}}_{\ell t} - \sum_{\ell|fr(\ell)=b} p^{\mathrm{L}}_{\ell t} = P^{\mathrm{D}}_{bt} + \sum_{s\in\mathcal{S}_b} p^{\mathrm{S,C}}_{st}; \forall b \in \mathcal{B}, \forall t \in \mathcal{T} \tag{18}$$

$$p^{\mathrm{L}}_{\ell t} = \frac{1}{X^{\mathrm{L}}_\ell} \left(\delta^{\mathrm{B}}_{fr(\ell)t} - \delta^{\mathrm{B}}_{to(\ell)t} \right); \forall \ell \in \mathcal{L} \setminus \tilde{\mathcal{L}}, \forall t \in \mathcal{T} \tag{19}$$

$$p^{\mathrm{L}}_{\ell t} = \frac{y^{\mathrm{L}}_\ell}{X^{\mathrm{L}}_\ell} \left(\delta^{\mathrm{B}}_{fr(\ell)t} - \delta^{\mathrm{B}}_{to(\ell)t} \right); \forall \ell \in \tilde{\mathcal{L}}, \forall t \in \mathcal{T} \tag{20}$$

$$-\overline{P}^{\mathrm{L}}_\ell \le p^{\mathrm{L}}_{\ell t} \le \overline{P}^{\mathrm{L}}_\ell; \forall \ell \in \mathcal{L}, \forall t \in \mathcal{T} \tag{21}$$

$$0 \le p^{\mathrm{G}}_{gt} \le \overline{P}^{\mathrm{G}}_g; \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \tag{22}$$

$$-R^{\mathrm{G,DN}}_g \Delta t \le p^{\mathrm{G}}_{gt} - p^{\mathrm{G}}_{gt-1} \le R^{\mathrm{G,UP}}_g \Delta t; \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \tag{23}$$

$$e^{\mathrm{S}}_{st} = e^{\mathrm{S}}_{st-1} + \left(\eta^{\mathrm{S,C}}_s p^{\mathrm{S,C}}_{st} - \frac{1}{\eta^{\mathrm{S,D}}_s} p^{\mathrm{S,D}}_{st} \right) \Delta t; \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \tag{24}$$

$$\underline{E}^{\mathrm{S}}_s \le e^{\mathrm{S}}_{st} \le \overline{E}^{\mathrm{S}}_s; \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \tag{25}$$

$$0 \le p^{\mathrm{S,D}}_{st} \le \overline{P}^{\mathrm{S,D}}_s z^{\mathrm{S}}_{st}; \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \tag{26}$$

$$0 \le p^{\mathrm{S,C}}_{st} \le \overline{P}^{\mathrm{S,C}}_s (1 - z^{\mathrm{S}}_{st}); \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \tag{27}$$

$$z^{\mathrm{S}}_{st} \in \{0,1\}; \forall s \in \mathcal{S}, \forall t \in \mathcal{T} \tag{28}$$

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where $fr(\ell)$ and $to(\ell)$ respectively index the sending- and receiving-end buses of line ℓ, whereas the duration of representative periods is denoted by Δt.

The optimization identifies the optimal values of the binary variables modeling line investments and storage operation, y_ℓ^{L} and z_{st}^{S}, as well as the optimal line power flows, $p_{\ell t}^{\mathrm{L}}$, generation levels, p_{gt}^{G}, charging and discharging power levels, $p_{st}^{\mathrm{S,C}}$ and $p_{st}^{\mathrm{S,D}}$, stored energy levels, e_{st}^{S}, and nodal voltage angles, δ_{bt}^{B}. The set of parameters comprises the nodal consumption levels, P_{bt}^{D}, the generation cost coefficients, $C_g^{\mathrm{G,P}}$, the maximum generation limits, $\overline{P}_g^{\mathrm{G}}$, the generation upward and downward ramp rates, $R_g^{\mathrm{G,UP}}$ and $R_g^{\mathrm{G,DN}}$, the charging and discharging cost coefficients, $C_s^{\mathrm{S,C}}$ and $C_s^{\mathrm{S,D}}$, the charging and discharging efficiency rates, $\eta_s^{\mathrm{S,C}}$ and $\eta_s^{\mathrm{S,D}}$, the minimum and maximum energy storage limits, $\underline{E}_s^{\mathrm{S}}$ and $\overline{E}_s^{\mathrm{S}}$, the rated charging and discharging power capacities, $\overline{P}_s^{\mathrm{S,C}}$ and $\overline{P}_s^{\mathrm{S,D}}$, the line investment cost coefficients, C_ℓ^{L}, the line series reactances, X_ℓ^{L}, and the line capacities, $\overline{P}_\ell^{\mathrm{L}}$.

The total cost minimized in (16) is the sum of line investment, generation, discharging, and charging costs. The binary nature of variables y_ℓ^{L} is expressed in (17). Constraints (18) are the power balance equations. The power flows through existing and candidate lines are modeled in (19) and (20), respectively, and bounded in (21). Generation limits and ramp rates are respectively set in (22) and (23). Finally, storage operation is characterized by (24)–(28), as done in (11)–(15).

According to [7], problem (16)–(28) can be simplified by dropping binary variables z_{st}^{S} for storage roundtrip efficiencies, i.e., $\eta_s^{\mathrm{S,D}}\eta_s^{\mathrm{S,C}}$, smaller than unity. However, as shown in the following case study, this condition is not sufficient. Let us consider a system with two disconnected buses, labeled as 1 and 2, where two generators and two loads, also designated as 1 and 2, are respectively located. Generation data are $C_1^{\mathrm{G,P}} =$ \$2/MWh, $C_2^{\mathrm{G,P}} =$ \$20/MWh, $\overline{P}_1^{\mathrm{G}} = 150$ MW, $\overline{P}_2^{\mathrm{G}} = 100$ MW, $R_1^{\mathrm{G,UP}} = R_1^{\mathrm{G,DN}} = 10$ MW/h, and $R_2^{\mathrm{G,UP}} = R_2^{\mathrm{G,DN}} = 100$ MW/h. A storage unit is also located at bus 1. As done in [7], the initial stored energy level is 0 MWh, $\eta_s^{\mathrm{S,D}} = 1.0$, and the charging and discharging costs are neglected, i.e., $C_s^{\mathrm{S,C}} = C_s^{\mathrm{S,D}} =$ \$0/MWh. Moreover, $\eta_s^{\mathrm{S,C}} = 0.9$, $\overline{P}_s^{\mathrm{S,C}} = \overline{P}_s^{\mathrm{S,D}} = 100$ MW, $\underline{E}_s^{\mathrm{S}} = 0$ MWh, and $\overline{E}_s^{\mathrm{S}} = 100$ MWh. In addition, a candidate 100-MW line is considered with $C_\ell^{\mathrm{L}} =$ \$5000 and $X_\ell^{\mathrm{L}} = 1$ p.u. on a base of 100 MVA and 138 kV. For illustration purposes, the operating condition consists of two hourly periods with both nodal demands respectively equal to 25 MW and 100 MW, and generators 1 and 2 initially producing 135 MW and 0 MW, respectively.

Problem (16)–(28) and its relaxed version have been both solved using the branch-and-cut algorithm of CPLEX for the two possible investment plans. All simulations were run to optimality by setting the optimality tolerance of CPLEX at 0%. Table II summarizes the optimal results. Note that the relaxed model gives rise to different optimal expansion and operational decisions incurring a lower total cost. However, this cost reduction is achieved by simultaneously charging and discharging the storage unit in period 2, which is not doable in practice. Moreover, an actually infeasible investment plan is turned into feasible. In other words, the optimal solution resulting from the relaxed model with both buses disconnected is not feasible for problem (16)–(28). These results demonstrate that the condition claimed in [7] is not sufficient to prevent simultaneous charging and discharging.

TABLE II
TRANSMISSION EXPANSION PLANNING – OPTIMAL RESULTS

	With z_{st}^{S}		Without z_{st}^{S}	
	Hour		Hour	
	1	2	1	2
p_{1t}^{G} (MW)	125.0	132.5	125.0	115.0
p_{2t}^{G} (MW)	0.0	0.0	25.0	100.0
$p_{st}^{\mathrm{S,C}}$ (MW)	75.0	0.0	100.0	50.0
$p_{st}^{\mathrm{S,D}}$ (MW)	0.0	67.5	0.0	35.0
e_{st}^{S} (MWh)	67.5	0.0	90.0	100.0
Installation of the candidate line	Yes		No	
Total cost (\$)	5515		2980	

V. CONCLUSION

This letter has disproved the recently reported criteria supporting the widely accepted use of a convex model for storage in MIP-based power system operation and planning. This is done by providing two counterexamples whereby it is shown that, unlike previously claimed, the convex model may not prevent impractical solutions featuring the simultaneous charging and discharging of storage units, which may lead to suboptimality and even infeasibility.

Further work will be devoted to exploring the existence of conditions under which the convex model holds. Another relevant avenue of research is the analysis of the practical techno-economic implications of the issue featured by the convex model.

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